

On-Chip Verified Quantum Computation

State-of-the-art and future directions

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Verified Quantum Computation

What is verification:

- Error bars / confidence intervals giving guarantees on computations.

Why is verification important:

- Quantum computation is noisy.
- Guarantees make the output more valuable.

How is verification done:

- That depends...



'Near Term' Verified Quantum Computation

Delegated Quantum
Computation + Benchmarking

Verified Universal Quantum Computation

‘Encrypted’ / ‘Encoded’
Quantum Computation + Prepare and
Send Qubits

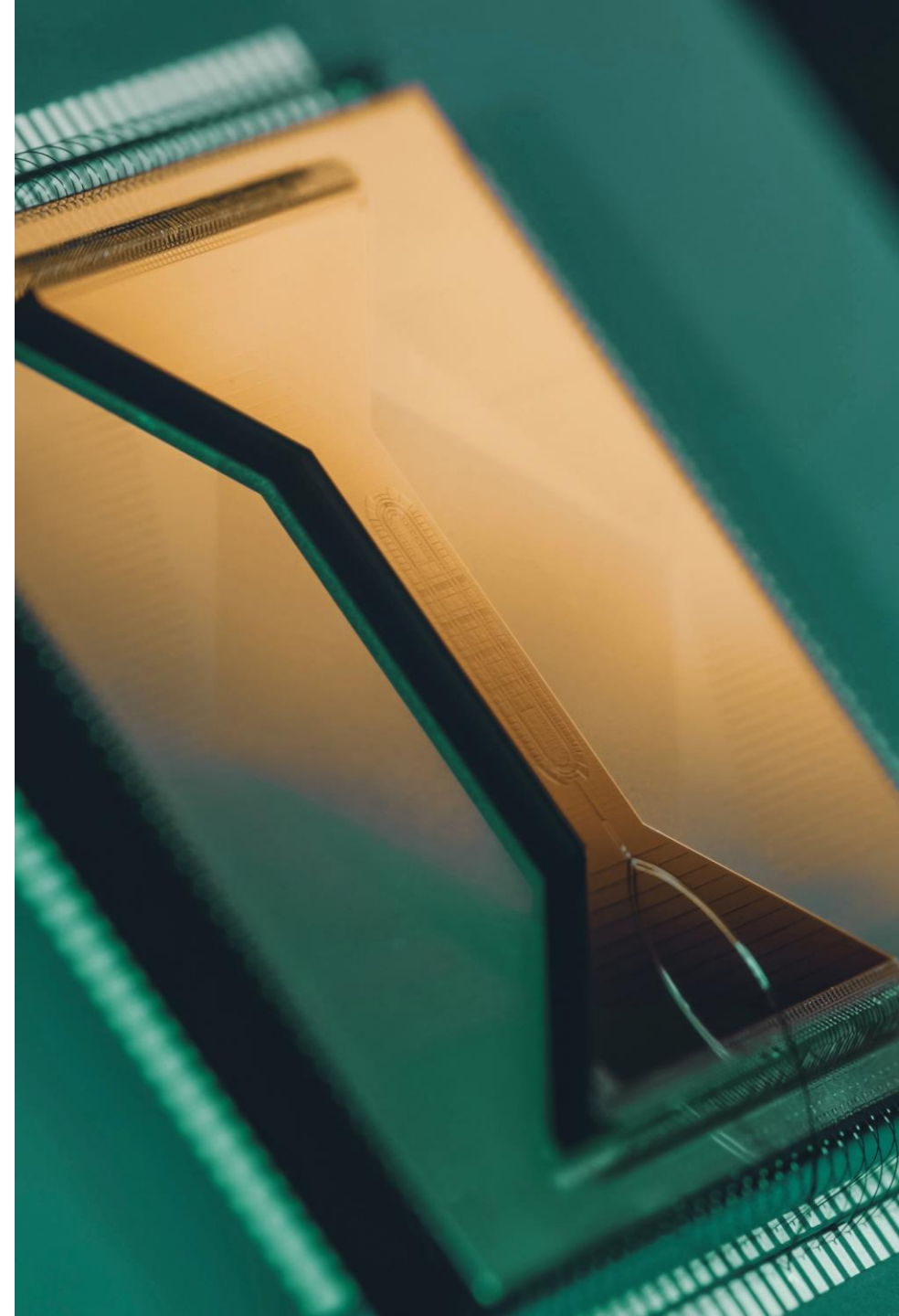
Practical Verified Computation

What we have:

- Benchmarking is not enough as computations become larger and more complicated.
- Preparing and sending qubits is too much.

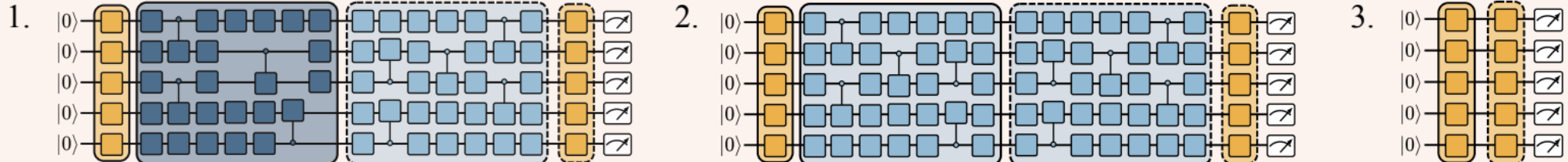
What is needed:

- Per circuit verification, so users have guarantees for their particular computation.
- Security against a malicious server is too much for most users, but they do care about noise.



Practical Verified Computation

b) Benchmarking circuits (mirror circuits with selective randomized compilation)

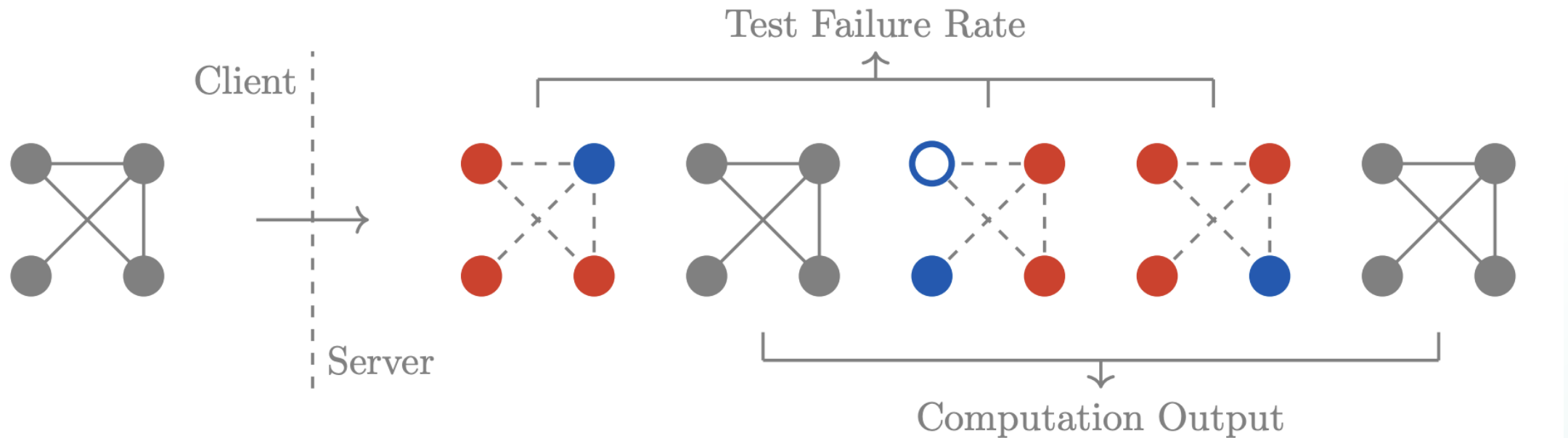


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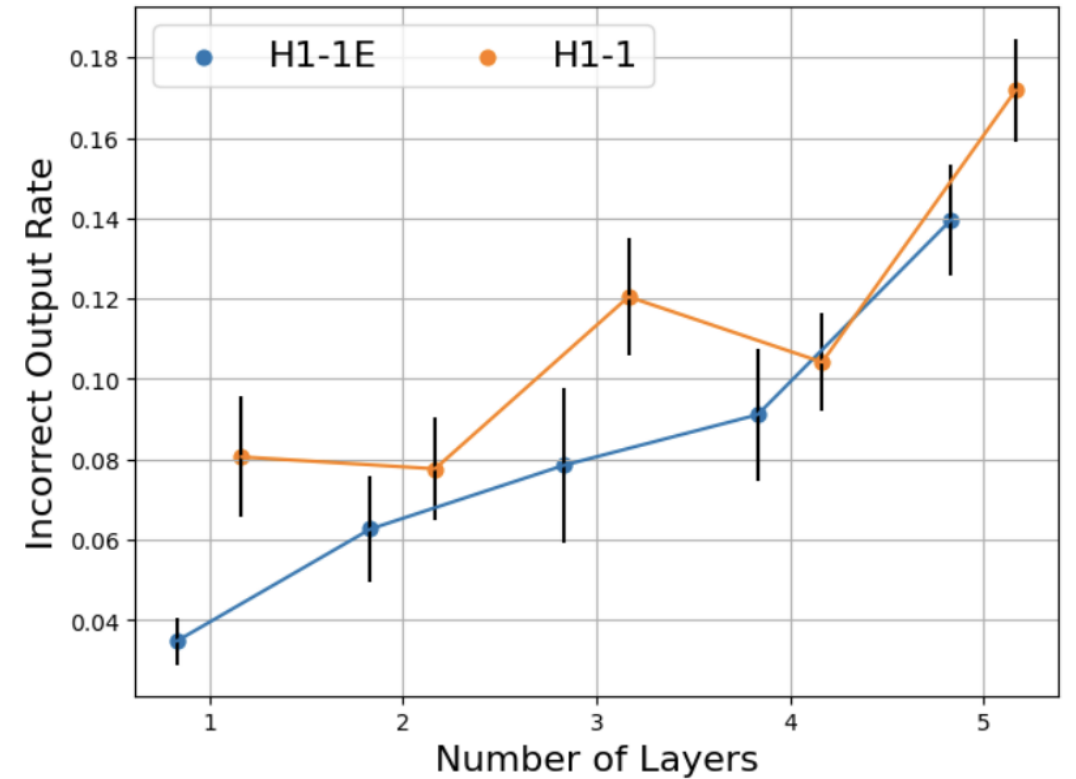
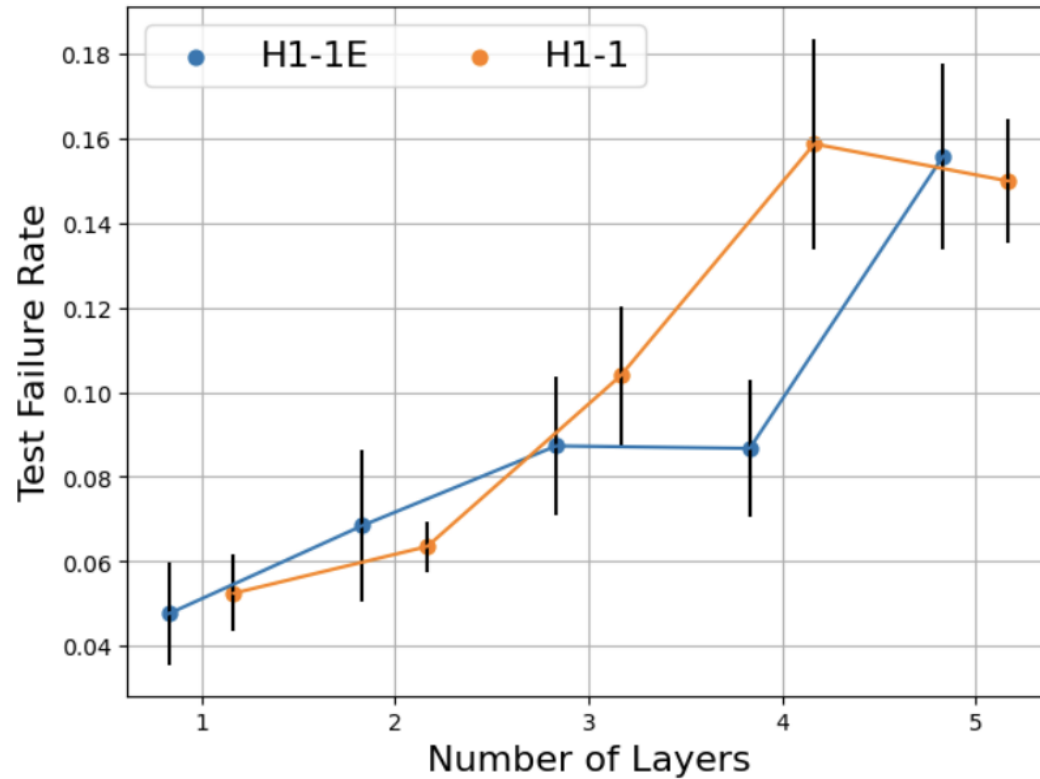
On-Chip Verified Quantum Computation

‘Encoded’ Quantum
Computation + Benchmarking

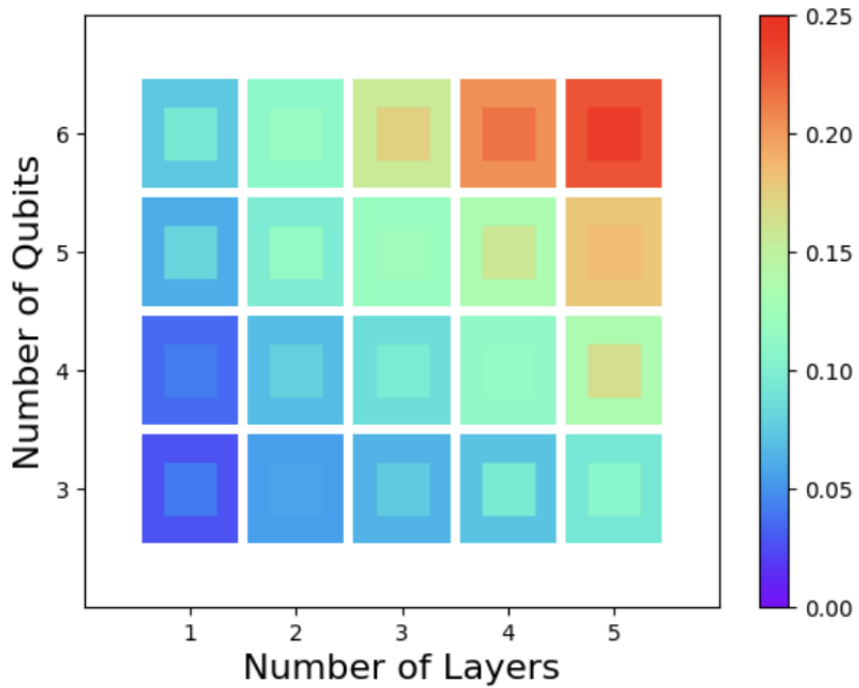
On-Chip Verified Quantum Computation



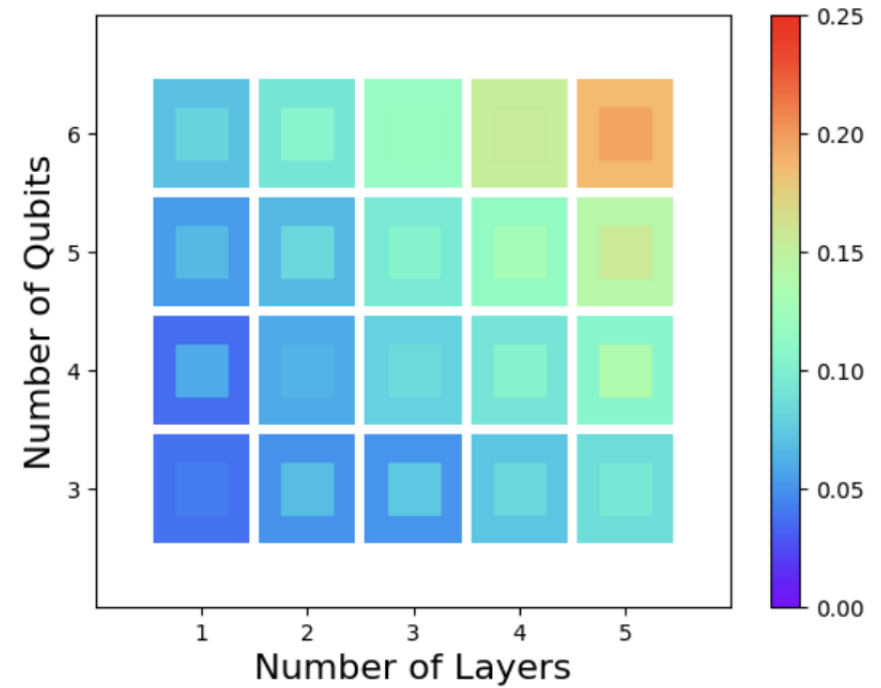
On-Chip Verified Quantum Computation



On-Chip Verified Quantum Computation



(a) Colour corresponds to the test failure rate



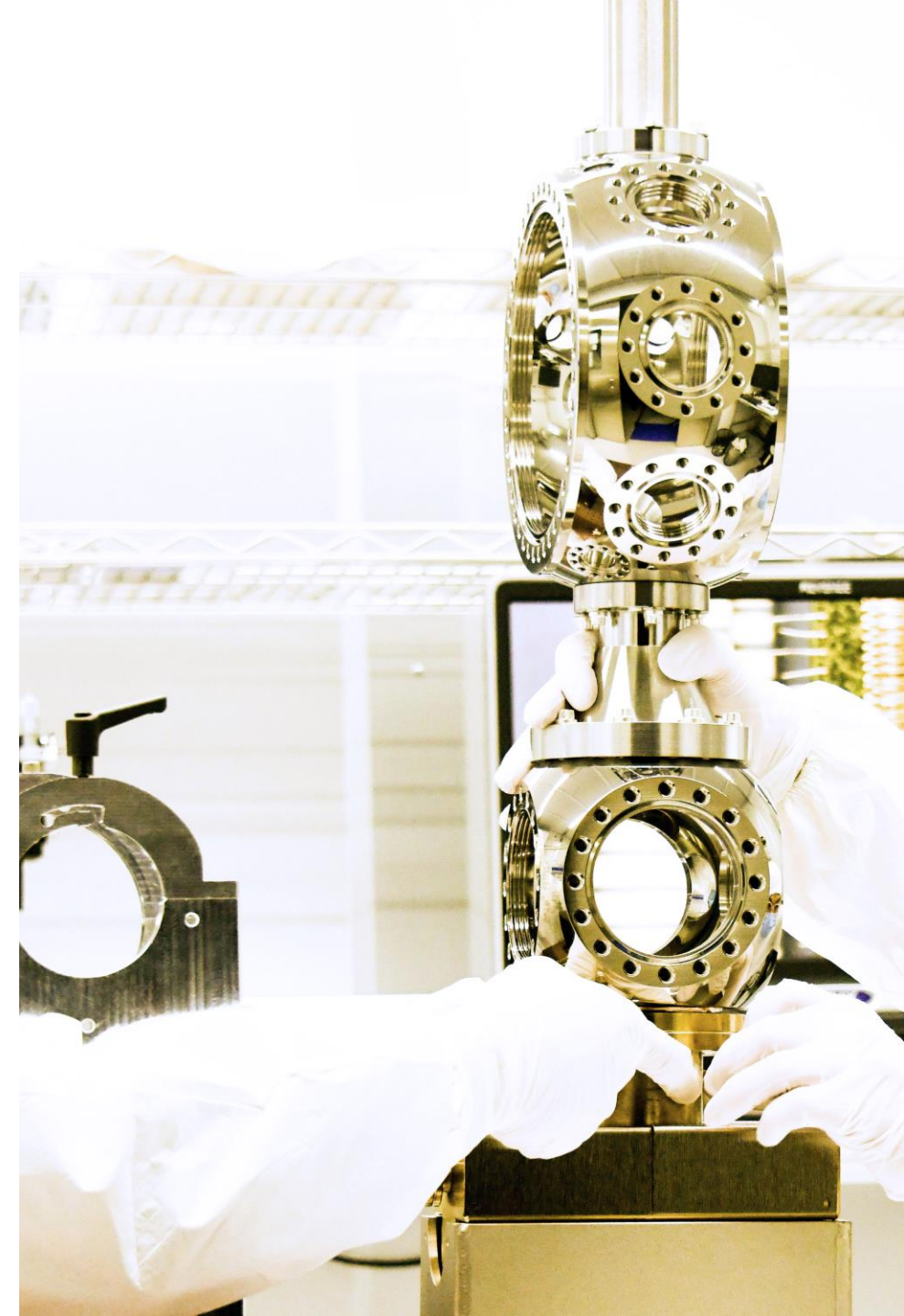
(b) Colour corresponds to incorrect output rate.

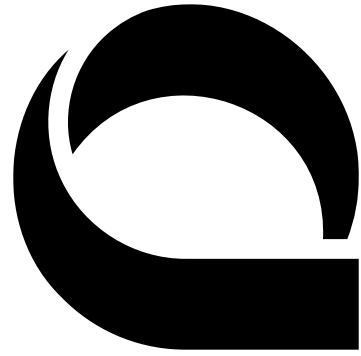
Future Directions

- Practical Verified Quantum Computation.
- Fault Tolerant Practical Verified Quantum Computation
- Practically Verified, Practical Quantum Computations Advantage? (RCS, Trotterized dynamic circuits, ...)



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QUANTINUUM